Deep Learning Based Medical Image Analysis For Early Disease Detection

Dataset & Results Analysis

Phase III project submission | 30 October 2026

Student / institution: enter the approved cover details. No identity or approval has been invented.

Dataset sources and scope

Official ISIC Challenge 2016 Task 3 release, CC0: https://challenge.isic-archive.com/data/ (retrieved and recorded in local metadata). The report names ISIC but does not specify a year/release; Task 3 binary benign/malignant was selected and is documented as an implementation choice.

The supplied Phase-I report also identifies NIH ChestX-ray14 and Kaggle APTOS Retinopathy. Those remain unavailable for local experiments as described below. The three tasks have separate labels and models; their data are not combined.

NIH ChestX-ray14 is not locally available because remaining workspace storage is insufficient for its full release alongside current files. APTOS 2019 requires Kaggle account sign-in and acceptance of competition rules; access is pending user download. Thus these two tasks are implemented in code but not trained/evaluated.

ISIC 2016 Task 3 dataset profile

Audit item	Observed value
Original listed records	1,279 (900 train + 379 official test)
Retained records	1,275 (896 train + 379 official test)
Labels	Benign 1,028; malignant 247
Training partition	896: benign 724; malignant 172
Official test partition	379: benign 304; malignant 75
Formats	JPEG
Most frequent dimensions	1024 x 768 (659 images)
Unreadable / missing references	0 / 0
Exact duplicate images	0 after removing 4 redundant training copies
Patient identifiers	Not supplied in Task 3 labels

Data quality and split policy

Four byte-identical training copies were identified; one repeated within training and three overlapped official test records. Labels matched. Redundant training copies were removed while all official test records were retained. No exact duplicates remain in the training/test manifest. Patient-level separation cannot be verified because patient/lesion IDs are absent; the source official train/test split is preserved and a stratified validation subset is drawn only from the official training partition.

The binary training set is imbalanced (724 benign, 172 malignant). Training-only class weighting is used. Image dimensions vary; preprocessing resizes to 224 x 224 and applies ImageNet normalization. Training augmentation is enabled; validation, test and inference use deterministic transforms.

Exploratory and model results

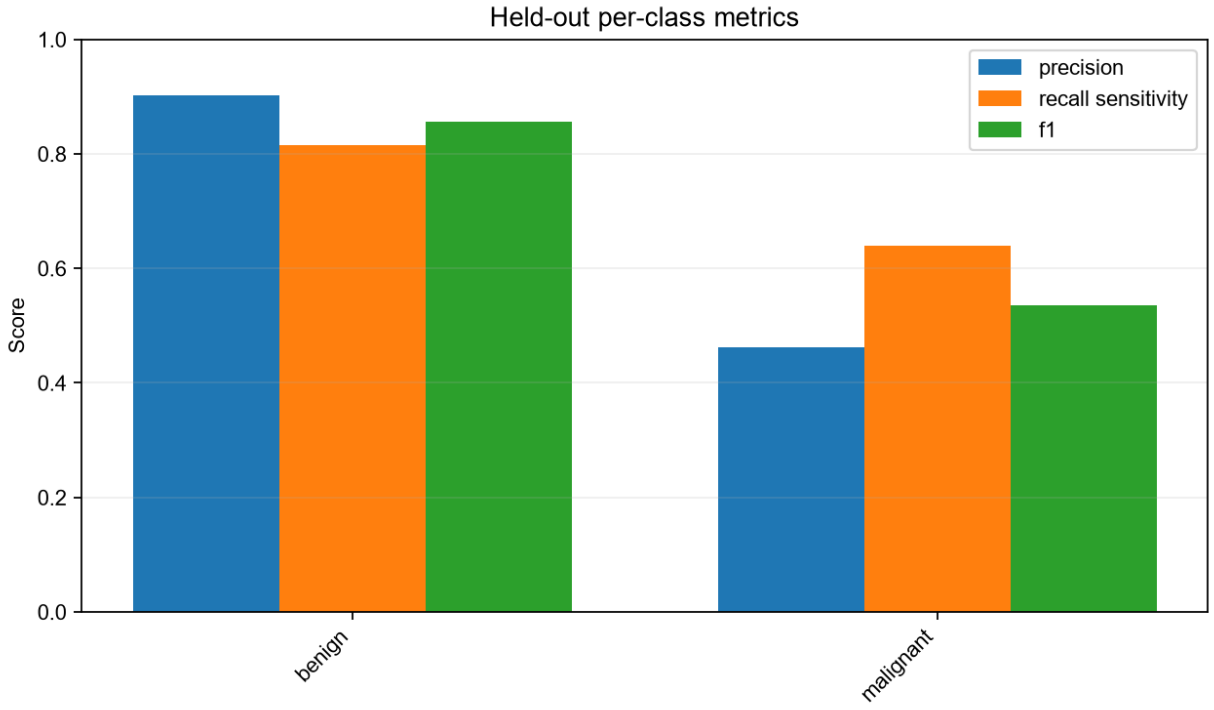

Figure 1. Class-wise metrics for the held-out official test partition.

Metric	Held-out official test result
Accuracy	0.7810
Macro precision	0.6817
Macro recall / sensitivity	0.7279
Macro F1-score	0.6965
Macro ROC-AUC (OVR)	0.8076

Confusion matrix (benign, malignant order): [[248, 56], [27, 48]]. The matrix reports both false positives and false negatives; see Performance Results for counts and curves.

Conclusions and limitations

Measured performance is specific to this release, split, preprocessing and single ResNet-18 run. It is not validated on other institutions or devices. No clinical claim is supported. NIH and APTOS findings cannot be reported until their actual files are supplied and audited.